

Hrithik Vasanthram

[LinkedIn](#) | [Github](#) | [Portfolio](#) | [LeetCode](#)

TECHNICAL SKILLS

Languages: Python, Java

Frontend: React.js, React Native, Next.js, HTML/CSS, Tailwind CSS

Backend: Node.js, Express.js, Socket.io, REST APIs, JWT, WebSockets

Databases: MongoDB, PostgreSQL

Tools / Frameworks / Libraries / Version Control / API Technologies: Git, GitHub Actions, Docker, AWS, Firebase, Figma, Bruno, Jest, crypto-js, DOMPurify

PERSONAL STATEMENT

Full-stack developer with a strong interest in building complete, production-quality products — from backend architecture to user-facing interfaces. I'm motivated by real-world impact and writing code that's maintainable, not just functional. Seeking an internship where I can learn directly from senior engineers and contribute to systems that real users depend on.

PROJECTS

HopePlates – Food Donation Platform [GitHub Repo](#) | [Live Demo](#)

Jan 2025 – Mar 2025

- **What you did:** Built a full-stack MERN platform connecting food donors, restaurants, and NGOs to reduce food waste, featuring role-based access across 5 user types and an admin verification pipeline.
- **Tech Stack & Frameworks:** MongoDB, Express.js, React, Node.js, Socket.io, JWT, Tailwind CSS, Multer, Firebase, Nominatim API.
- **Problem Solved / Use Case:** Donors and restaurants had no efficient way to connect with NGOs for food redistribution — this platform automates the entire workflow from listing to pickup confirmation.
- **Challenges Tackled / Learning:** Designing a permission system across 5 user types required careful middleware architecture; learned how to structure scalable route protection using JWT and Express.

Onyx – End-to-End Encrypted Chat Rooms [GitHub Repo](#) | [Live Demo](#)

Apr 2025 – May 2025

- **What you did:** Built a real-time MERN chat app with client-side AES-256 encryption via crypto-js — the server stores only ciphertext, never plaintext, designed for privacy-conscious secure group communication.
- **Tech Stack & Frameworks:** MongoDB, Express.js, React 19, Node.js, Socket.io, crypto-js, Firebase Auth, DOMPurify, marked, Groq API.
- **Problem Solved / Use Case:** Standard chat apps store readable messages server-side — Onyx solves this with three room modes: Normal (persisted), Ghost (auto-purged on disconnect), and Couples (2-user cap).
- **Impact / Usage:** Deployed and live; features include read receipts, typing indicators, QR-based invites, and a Groq LLaMA 3.3 70B AI assistant (Cipher) accessible via @cipher commands.
- **Challenges Tackled / Learning:** Implementing conditional room destruction with MongoDB purge logic on Socket.io disconnect events required careful async handling; gained deep expertise in real-time state management across concurrent clients.

EDUCATION ENROLLMENT

Kalvium's UG Program in CS (Software Product Engineering)

2024 – 2028

Campus: Punjab | Bachelor's Enrollment: B.Tech, Lovely Professional University

ACHIEVEMENTS & EXTRACURRICULARS

Competitions / Leadership

- Finalist – KVS Jigyasa Innovation Challenge 2023, Chennai.
- Runner-Up – One India 2025, Lovely Professional University.

Technical Contributions

- Merged 3 PRs into [slidevjs/slidev](#) (46k ★) — [View all commits](#).
- Solved 130+ DSA problems on LeetCode across arrays, graphs, and dynamic programming — consistently practising algorithmic problem-solving.